

Budget Constraint

JEB104 Microeconomics I
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Basic Definitions

Consumption bundle:

$$\mathbf{x} = (x_1, x_2, x_3, \dots, x_n)$$

$x_i, i = 1, \dots, n$: amount of i -th good/commodity

- 1 $x_i \geq 0$
- 2 perfectly divisible $x_i \in \mathbb{R}_{+,0}$

Consumption set:

set of all potential consumption bundles

$$X = \mathbb{R}_{+,0}^n = \{\mathbf{x} \in \mathbb{R}^n : x_i \geq 0 \text{ for } i = 1, \dots, n\}$$

- convex

If $\mathbf{x}_a, \mathbf{x}_b \in X$, then $\mathbf{x}_c = \lambda \mathbf{x}_a + (1 - \lambda) \mathbf{x}_b$ is also in X .

Basic Definitions

Consumer limited by $m \geq 0$ – income, the amount of money to spend

$\mathbf{p} = (p_1, \dots, p_n)$ is vector of prices, p_i is price of x_i for all x_i , $i = 1, \dots, n$

Budget constraint:

$$\sum_{i=1}^n p_i x_i \leq m$$

Budget set:

affordable consumption bundles at prices $\mathbf{p}$

$$B = \{\mathbf{x} \in \mathbb{R}_{+,0}^n : \sum_{i=1}^n p_i x_i \leq m\}$$

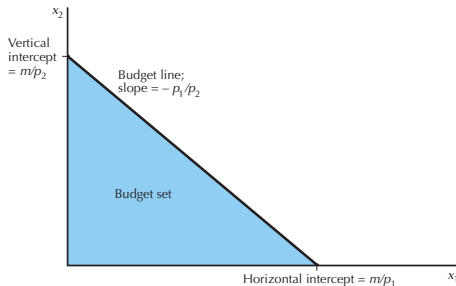
- 1 nonempty = at least 0 is in the set
- 2 closed = contains its own boundary
- 3 bounded
- 4 convex
- 5 $0 \in B$

2-commodities Case

Simplification for graphical representation: 2 commodities

$$p_1x_1 + p_2x_2 \leq m$$

Budget line: $p_1x_1 + p_2x_2 = m$



Slope $-\frac{p_1}{p_2}$

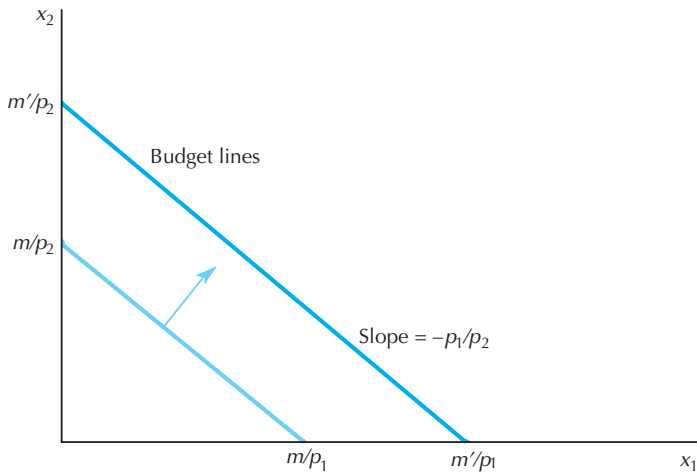
- rate at which it is possible to substitute x_1 for x_2 in the market (1 unit of x_1 can be added when removing $\frac{p_1}{p_2}$ units of x_2)
- opportunity cost of consuming x_1

2-commodities case too restrictive?

x_1 and **composite** good x_2 : the amount of money spend on all other goods than x_1

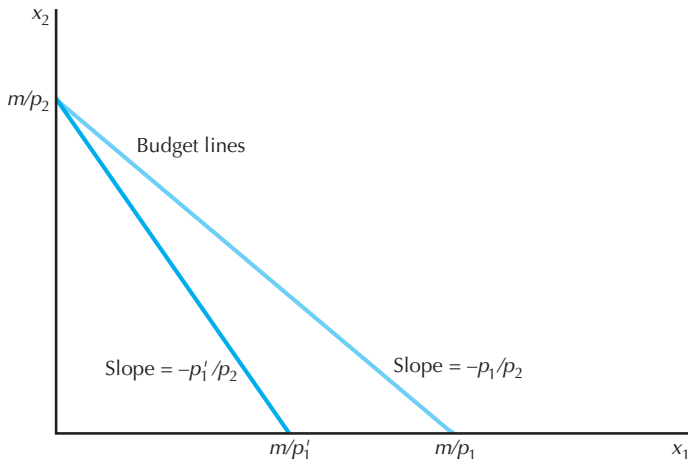
Increasing Income

Parallel shift outward of the budget line



Increasing Price of Good 1

Budget line becomes steeper without a change of the vertical intercept



What happens if both prices double? And what if the income doubles as well?

Numeraire

3 parameters influencing the budget set: p_1, p_2, m

Use just 2 parameters to describe the same budget set:

x_2 is numeraire $\rightarrow p_2 = 1$:

$$\frac{p_1}{p_2}x_1 + x_2 = \frac{m}{p_2} \rightarrow p'_1x_1 + x_2 = m'$$

$m = 1$:

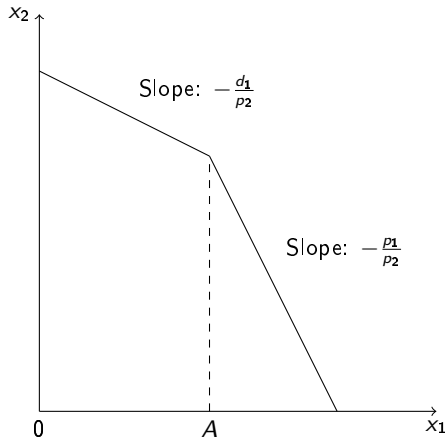
$$\frac{p_1}{m}x_1 + \frac{p_2}{m}x_2 = 1 \rightarrow p''_1x_1 + p''_2x_2 = 1$$

Some Special Cases

- Kinky budgets
 - e.g. a certain discount on one good, however, limited by a maximum number of discounted units
- (Progressive) tax
 - Consumption above certain level is taxed more than below
- Rationing
 - Only allowed to purchase some maximum number of a good

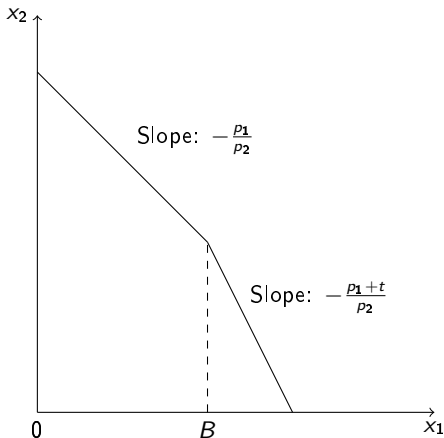
Kinky Budgets

Good 1 is sold for a lower price d_1 , however, only A units get the discount:



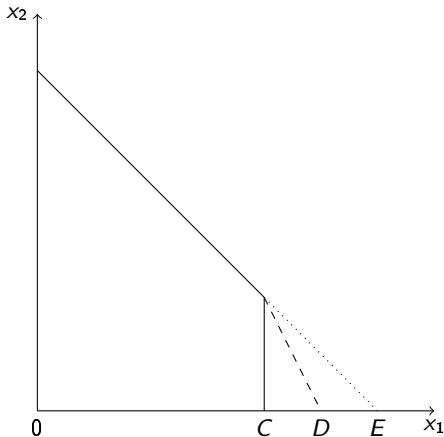
Progressive Tax

First B units of good 1 are not taxed, additional units are taxed by a tax t .



Rationing

Consumer only allowed to buy a maximum of C units of good 1:



Point E : unregulated market, Point D : black market